

MULTIPLECORP

Company Management System

Scope, Golden Business Flow, and Definition of Done

Astra Planning Reference / Project Memory

Purpose. This document captures the current understood product scope, end-to-end operational flow, and completion criteria for MultipleCorp. It is intended to give Astra a stable planning reference and enough context to identify better implementation approaches without losing Owner intent.

Authority rule. If this reference conflicts with a later explicit Owner decision or an approved/locked canonical repository specification, the latest approved source wins. Astra should flag contradictions rather than silently choosing.

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1. Executive Understanding

MultipleCorp is not a cut-down MVP. The intended V1 includes the full set of previously requested and approved business capabilities, except items the Owner has explicitly removed or deferred. The 18-day build window is an execution constraint, not permission to silently reduce product scope.

Product model. MultipleCorp is a mobile-first, multi-company internal business management web application centered around Project as the operational hub. It connects quotation, procurement, inventory/drop-ship, delivery, documents, billing, payment, profitability, reporting, auditability, permissions, and recovery into one coherent workflow.

Primary users. The built-in launch roles are Super Admin / Owner and Admin Operasional. The authorization model must remain capability/permission based so the Owner can create additional roles later without architecture refactoring.

User-facing language and UX. All end-user UI is Bahasa Indonesia, using simple and natural business terminology. The product is mobile-first while desktop is optimized for intensive operational productivity. shadcn/ui is the primary UI system. The visual direction is professional, modern, elegant, calm, efficient, information-dense, and explicitly anti-AI-slop.

2. Scope Principles

- **Full approved scope stays in V1:** No feature or business logic is removed only because the schedule is aggressive.
- **Explicit cuts stay cut:** Supplier Comparison is removed from V1. Direct SIPLAH API integration is deferred; SIPLAH remains a transaction channel. Full formal accounting is not the current finance model.
- **Project-centered operation:** The Project page is the main operational workspace linking customer, items/services, quotation, purchases, fulfillment, documents, billing, payments, and activity history.
- **One shared physical warehouse:** Physical stock is shared, while stock movement and cost/source attribution remain traceable to the relevant company/project/source.
- **Corrections preserve history:** Important business records are not silently deleted or overwritten. Use revision, void, reversal, return, payment reallocation, or compensating transactions with reason and audit trail.
- **Security and data integrity are product features:** Authorization, company isolation, validation, idempotency, concurrency safety, N+1 prevention, auditability, backup, and restore are required completion criteria, not optional hardening.

3. V1 Product Scope

The following capability map is the current understood V1 scope. It is deliberately broad because the Owner does not want requirements silently reduced for schedule convenience.

#	Capability	Required scope
1	Authentication	Login, logout, password reset, secure session, password hashing, login throttling.
2	Users & built-in roles	Super Admin / Owner and Admin Operasional.

#	Capability	Required scope
3	Dynamic roles & permissions	Owner can later create custom roles and assign capabilities/permissions.
4	Company access	Admin can be restricted to selected companies; Owner sees all.
5	Multi-company	Support existing companies and future CV/PT/legal entities added by Owner.
6	Company identity	Legal identity, NPWP, bank account, address, logo, stamp/signature, director, document identity.
7	Owner dashboard	Consolidated and per-company operational/financial visibility.
8	Admin dashboard	Action-focused operational queue: projects, receiving, delivery, invoices, admin docs, overdue, low stock.
9	Client master	All clients use one generic organization model; UNY/UGM are normal clients.
10	Client hierarchy	Organization -> Unit/Department -> PIC/contact; unit-specific addresses where needed.
11	Supplier master	Supplier records, purchase history, product history, previous/last purchase price.
12	Products & services	Products/services, internal SKU, units, images, active/inactive status.
13	Barcode	Manufacturer barcode, generated internal barcode, unknown barcode handling, duplicate protection.
14	Serial tracking	Optional serial-number tracking for electronics/selected products.
15	Pricing	Purchase price, selling price, SPJ/reference price, transaction snapshots and price history.
16	Tax/adjustments	Configurable tax/fee/VA/adjustment components; avoid unsafe hard-coded assumptions.
17	Project core	Project number, company, client/unit/PIC, items/services, status, deadlines and operational history.
18	Quotation	Generate, revise, version, PDF/print, validity, statuses, approval.
19	Purchase	Supplier selection, actual purchase price, shipping/tax components, company/project reference.
20	Optional PO	PO is not mandatory; multiple POs may exist where procurement uses multiple suppliers.

#	Capability	Required scope
21	Shared warehouse	One physical warehouse for all companies.
22	Stock/cost attribution	Physical stock shared while source/cost-owning company attribution remains traceable.
23	Storage location	Rack/location information for stored products.
24	Goods receiving	Barcode scan, partial receiving, serials, damaged goods, receiving evidence.
25	Goods dispatch	Barcode scan, project/client reason, user log, safe reversal, no history deletion.
26	Stock ledger	Auditable stock movement ledger plus optimized current balance.
27	Stock opname	Physical scan/count, variance, controlled adjustment with reason/audit.
28	Minimum stock/restock	Per-product minimum stock, low-stock warning and restock suggestion.
29	Returns/reversal	Purchase/sales return and reversal/correction flows.
30	Drop-ship	Supplier-to-client fulfillment without pretending goods entered the warehouse.
31	Delivery	Partial/full delivery, delivery documents, proof/signature/name/condition uploads.
32	Document engine	Reusable document generation framework with per-company templates/identity.
33	Generated documents	Quotation, PO, delivery note/faktur pengiriman, nota, kuitansi, receipt attachment, invoice, BAST and other company-issued templates.
34	External/supporting documents	SPK/client order, official tax invoice, NPWP/NIB, bank statement, HPS and other third-party documents via upload/reference.
35	Document versioning	Draft, issued/final, revision, void/cancel where relevant; preserve prior versions.
36	Document numbering	Automatic, company/type/period aware, concurrency-safe unique numbering.
37	Administrative documents / SPJ	Flexible per-project/client checklist; owner/admin selects required documents.
38	SIPLAH channel	Regular vs SIPLAH transaction channel, SIPLAH metadata, future API-ready boundary.
39	Billing	Separate invoice document from billed status/date/due date.

#	Capability	Required scope
40	Receivables	Unbilled/billed/unpaid/partial/paid/overdue statuses and aging.
41	Payments	Full/partial/DP/termin when applicable, method, company bank, reference and proof.
42	Overpayment	Customer credit or refund/correction; never silently treated as profit.
43	Payment correction	Reallocation/correction with permission and audit trail.
44	Managerial finance	Revenue, direct cost/HPP, project expenses, cash in/out, receivables, profitability.
45	Profitability	Profit/margin per project, company, period and consolidated group.
46	Reports	Date/company filters for stock, purchases, sales, AR, payments, profit and managerial cashflow.
47	Export/print	Excel/PDF/print where operationally relevant.
48	Notifications	In-app/dashboard alerts for low stock, overdue, incomplete work and actions requiring attention.
49	Audit log	Critical changes: stock, price, invoice, payment, permissions, void/correction, documents.
50	Search	Fast search for product/SKU/barcode/project/client/document.
51	Data migration	Products, prices, suppliers, clients, opening stock, open work/receivables and useful legacy records.
52	Indonesian UI	All user-facing UI uses clear, natural Bahasa Indonesia.
53	Responsive UX	Mobile-first; desktop optimized for high-intensity administration.
54	UI system	React + TypeScript + Inertia + shadcn/ui + Tailwind.
55	Anti AI-slop design	Operational, elegant and modern without generic SaaS/AI visual clichés.
56	Security	Authorization, company isolation, validation, CSRF, cookies, XSS, SQL injection, clickjacking, upload security, IDOR/BOLA.
57	Data integrity	Transactions, row locking/atomicity where needed, database constraints, safe correction semantics.

#	Capability	Required scope
58	Idempotency	Duplicate payment/receiving/dispatch/document issuance must not duplicate state.
59	Performance	No N+1, pagination, indexing, efficient search, no unbounded collection loads.
60	Backup & recovery	Database + document backups, offsite copy, restore procedure/test, current target RPO <=24h, RTO <=4h.
61	Testing	Pest/PHPUnit + TestSprite + CI + performance/load testing where justified.
62	Production operations	Existing VPS, near-zero incremental recurring cost, self-hosted/open-source oriented deployment.

4. Current Exclusions / Deferrals and Their Nuance

- **Supplier Comparison:** Explicitly removed from V1 by the Owner. Do not reintroduce it.
- **Direct SIPLAH API integration:** Deferred. V1 keeps SIPLAH as a transaction channel and should be integration-ready.
- **Full formal accounting / General Ledger:** Not the current finance model. V1 uses managerial finance, not a full accounting engine.
- **Native Android/iOS application:** Not part of the current requested webapp delivery; the webapp itself must be mobile-first.
- **AI runtime features:** Not currently requested as a core MultipleCorp capability.

5. Golden Business Flow - Login to Completion

LOGIN

- > Authentication
- > Company Access + Permission Check
- > Dashboard
- > Create / Open Project
- > Select Company
- > Select Client
- > Select Unit / Department / PIC
- > Select Transaction Channel (Regular / SIPLAH)
- > Add Products & Services
- > Set Price / Tax / Transaction Values
- > Create Quotation
- > Revise if needed
- > Quotation Approved
- > Determine Fulfillment Need
 - |-- Stock Available -> Allocate / Continue
 - |-- Purchase Needed -> Select Supplier -> Purchase -> PO? (Optional)
- > Fulfillment Method
 - |-- Warehouse -> Receive / Scan / Serial / Balance Update
 - |-- Drop-Ship -> Supplier-to-Client Confirmation
- > Delivery (Partial or Full)

- > Delivery Note / Faktur Pengiriman
- > Proof of Handover / Receipt
- > Transaction Documents (Nota / Kuitansi / BAST as required)
- > Invoice
- > Administrative Documents / SPJ Requirements
- > Mark as Billed
- > Due Date / Receivable Tracking
- > Payment
 - | -- Partial -> Remaining Receivable -> Additional Payment
 - | -- Full -> Paid
- > Completion Validation
- > Project Complete
- > Dashboard & Reports Update

6. Detailed Stage Behavior

6.1 Login and authorization

Authenticate securely; establish session; then enforce role/capability, company scope, resource authorization, and business-state preconditions. Owner has full company visibility; Admin sees only explicitly allowed companies.

6.2 Dashboard

Owner dashboard emphasizes consolidated/per-company business visibility. Admin dashboard emphasizes next operational actions: quotations, purchases/POs, receiving, delivery, invoices, admin documents, overdue receivables, and low stock.

6.3 Project workspace

Project is the single operational workspace. It should expose overview, items/services, quotation, purchasing, fulfillment, delivery, documents, administration, billing, payment, and activity/history without forcing repeated data entry.

6.4 Quotation

Generate from project data; support draft, sent, revision, approval and version history. Approved quotation transitions the project into execution without retyping the same data.

6.5 Procurement

If stock is insufficient, choose supplier directly, record actual purchase price and related cost/tax components, and create PO only when needed. Supplier Comparison is not part of V1.

6.6 Fulfillment

Two valid paths: warehouse or drop-ship. Warehouse receiving updates stock only when physical receipt is confirmed. Drop-ship records fulfillment without fabricating warehouse movement.

6.7 Inventory

Receiving and dispatch are auditable movements, not manual stock number edits. Dispatch validates project, stock availability and permissions; race conditions must never create negative stock. Stock opname uses controlled adjustment with reason.

6.8 Delivery

Support partial and full delivery. Each delivery has its own history and supporting evidence. Generate the appropriate delivery document and allow proof/signature uploads.

6.9 Documents

Owner/Admin selects required generated or uploaded documents. Company-issued documents can be generated; third-party/official external documents are uploaded/referenced. Final documents are versioned/locked and corrected by revision/void rather than silent overwrite.

6.10 Invoice and billing

Invoice is a document; billing is the operational state that the invoice has been submitted for payment. Track billing date, due date and receivable status.

6.11 Payment

Record date, amount, method, destination company bank, reference and proof. Partial payment leaves a receivable balance. Overpayment becomes credit/refund handling, not profit. Reallocation/correction requires audit.

6.12 Completion

Project completion is gated by fulfillment, required documents, billing/payment resolution, open corrections/returns, and audit completeness - not merely by delivery.

7. Project Completion Gate

Completion condition	Required	Interpretation
Goods/services fulfilled	YES	All required fulfillment completed, or remaining quantity formally cancelled/closed.
Delivery resolved	YES	All delivery records complete and no unresolved delivery discrepancy remains.
Returns/corrections resolved	YES	No open return, reversal or correction that materially affects project state.
Required documents issued	YES	Invoice and other required company-generated documents are issued/final where applicable.
Administrative requirements resolved	YES	Required SPJ/admin checklist items are complete, uploaded, or formally waived with reason when allowed.
Billing resolved	YES	Project has been marked billed when billing is required.
Receivable resolved	YES	Receivable is zero or has an explicitly approved resolution path.
Payment correction resolved	YES	No unresolved payment reallocation/correction remains.
Inventory state resolved	YES	No incomplete stock transaction or pending stock correction remains.

Completion condition	Required	Interpretation
Audit trail complete	YES	Critical actions and corrections remain traceable.

8. Feature Definition of Done - Eight Mandatory Gates

Gate	What must be proven	Status to call DONE
1. Business	All business rules and acceptance criteria are satisfied, including non-happy paths that matter.	PASS
2. UI/UX	Bahasa Indonesia, mobile-first, desktop professional, shadow-based, accessible enough for real use, with loading/error/empty states.	PASS
3. Backend	Correct action/use-case behavior, server-side validation, explicit error handling, thin controllers/components.	PASS
4. Database	Migrations, foreign keys, unique/check/not-null constraints, indexes, safe money representation and data model integrity are correct.	PASS
5. Security	Authentication/authorization, company isolation, IDOR/BOLA protection, CSRF/session/file security and relevant controls are verified.	PASS
6. Integrity	Concurrency, locking/atomicity, idempotency, transaction boundaries, correction semantics and auditability are correct where applicable.	PASS
7. Performance	No N+1 regression, appropriate pagination/indexing, no unbounded loads, reasonable query behavior.	PASS
8. Verification	Automated backend tests pass, relevant TestSprite flow passes, CI passes, and business-facing features receive UAT where needed.	PASS

9. Concrete Definition-of-Done Examples

9.1 Goods Dispatch

- Authorized user and correct company scope.
- Valid barcode/product/project and quantity.
- Sufficient stock.
- Duplicate submit cannot duplicate movement.
- Two simultaneous dispatches cannot make stock negative.
- Stock movement and balance remain consistent.

- Audit event is recorded.
- Reversal/correction path exists.
- No N+1 or unbounded queries.
- Mobile flow is usable.
- Backend + concurrency + TestSprite + CI checks pass.

9.2 Payment

- Normal payment, partial payment and full payment work.
- Invalid/negative/zero amounts are rejected.
- Unauthorized/wrong-company payment is rejected.
- Duplicate request/idempotency key cannot create duplicate payment.
- Overpayment produces approved credit/refund behavior, not profit.
- Payment reallocation/correction is audited.
- Receivable balance remains correct.
- Proof upload follows file/security rules.
- Mobile and desktop UI are usable.
- Backend, integration, TestSprite and CI checks pass.

9.3 Invoice

- Generated from correct project/client/company snapshot.
- Correct quantities and transaction prices.
- Company identity/template is correct.
- Number generation is concurrency-safe and unique.
- PDF/print output is correct.
- Issued invoice is locked from silent edits.
- Revision preserves previous version.
- Void/cancel preserves history and reason.
- Permission/company isolation is enforced.
- Automated tests pass.

10. Build Unit Completion Checklist

BU STATUS

Business Acceptance	PASS
UI	PASS
Responsive	PASS
Authorization	PASS
Validation	PASS
Company Isolation	PASS
Database Constraints	PASS
Race Condition Review	PASS / N/A + reason
Idempotency Review	PASS / N/A + reason
N+1 Review	PASS
Audit	PASS / N/A + reason
Pest/PHPUnit Tests	PASS
Static Analysis	PASS
Frontend Typecheck	PASS
TestSprite	PASS / N/A + reason

Regression	PASS
Documentation	UPDATED
CURRENT_STATE	UPDATED

FINAL: DONE

Not acceptable as completion evidence: “looks good”, “feature implemented”, or “UI works”.
Completion requires machine-verifiable and business-verifiable evidence appropriate to the feature.

11. When a Module Is Done

A module is complete only when all of its required capabilities and cross-cutting protections work together. Example - Inventory is not complete when a stock page exists; it requires the coherent set below:

- Product/barcode identity
- Goods receiving
- Current stock balance
- Stock movement ledger
- Goods dispatch
- Adjustment
- Stock opname
- Return/reversal
- Drop-ship semantics
- Company/cost attribution
- Authorization/company isolation
- Concurrency/idempotency protections
- Automated tests and regression

12. When MultipleCorp V1 Is Release-Done

Release gate	Required status	Meaning
Feature traceability	PASS	Every V1_REQUIRED capability maps to a Build Unit and acceptance criteria.
Build completion	PASS	All mandatory Build Units are DONE under the defined gates.
Golden Flow	PASS	End-to-end Login -> Project -> Quotation -> Purchase -> Receiving/Drop-Ship -> Delivery -> Invoice -> Billing -> Payment -> Complete passes.
Owner UAT	PASS	PASS.
Admin Operasional UAT	PASS	PASS.
Migration validation	PASS	PASS for agreed migrated data.
Permission/company isolation	PASS	PASS.
Inventory integrity	PASS	PASS.
Payment integrity	PASS	PASS.
Concurrency/idempotency critical tests	PASS	PASS.

Release gate	Required status	Meaning
TestSprite regression	PASS	PASS.
CI	PASS	PASS.
Security checks	PASS	PASS.
N+1/performance review	PASS	PASS.
Backup	PASS	Successful and verifiable.
Restore procedure	PASS	Verified against the defined recovery approach.
Production configuration	PASS	Validated; debug off; secrets and access correct.
Deployment	PASS	Successful.
Post-deploy smoke test	PASS	PASS.

13. Planning Guidance for Astra

- **Use this as memory, not a substitute for canonical repository truth:** Astra should incorporate this understanding into canonical P1-P11 documents rather than relying on this PDF forever.
- **Preserve capability, simplify implementation:** Astra is encouraged to find an easier build strategy as long as the user-visible/business capability is not silently removed.
- **Find gaps proactively:** Look for missing invariants, concurrency risks, unclear correction rules, permission leakage, migration/recovery gaps, and operational friction.
- **Escalate real Owner decisions:** If a proposed change affects business workflow, money meaning, stock ownership, permissions, recurring cost, or removes requested capability, mark OWNER_DECISION_REQUIRED.
- **Avoid overengineering:** Use the simplest safe architecture that preserves the required outcome. Do not turn the broad feature set into needless infrastructure complexity.
- **Build for continuity:** Planning and Build Units should allow Claude Opus 5.5 or another future executor to implement without needing historic chat context.

14. Core Completion Principle

A feature is not complete because it exists on screen. It is complete when it is correct for the business, secure against unauthorized or malicious use, consistent under concurrent/repeated requests, performant enough for real data, auditable, automatically tested, and usable by Owner/Admin in real operations.

Planning reference only - latest explicit Owner decisions and approved/locked repository specifications take precedence.